Even subsets of a polygon and an equal-label presentation

Abstract

The halving construction of Montero and Ivić Weiss, applied to a polygon, gives a self-dual regular map of type $\{n, n\}$ with $n2^n$ flags. We identify its distinguished generators in the semidirect product of the even subsets of an n-cycle by the dihedral group. This recovers the main construction of Li and Zhang from the previously established polygon corollary. We then prove their exact universal presentation by a normal subgroup generated by commuting involutions. The proof works for every $n \geq 3$, without enumerating groups. The existence theorem and presentation are attributed to their sources; the contribution here is their explicit identification within the common construction and a checked algebraic part of the presentation argument.

1. The earlier construction and the source contract

Write D_{2n} for the dihedral group of order $2n$. A marked group $G = \langle r_0, r_1, r_2 \rangle$ is a string C-group if its generators are nonidentity involutions, $r_0 r_2 = r_2 r_0$, and

$$G_I \cap G_J = G_{I \cap J}, \qquad G_I = \langle r_i : i \in I \rangle, \qquad I, J \subseteq \{0, 1, 2\}. \tag{1}$$

We retain the regular-polytope correspondence of McMullen and Schulte: such a group is the full flag-regular automorphism group of a regular abstract polyhedron, of type $\{|r_0 r_1|, |r_1 r_2|\}$. Reversal of the marking by an automorphism gives a duality of that polyhedron.

The polygon case of Montero–Ivić Weiss [MW20, Section 4, physical page 9] is already recovered in the common manuscript [C, Corollary C.Polygon]. For an n-gon, it gives a self-dual orientable regular map with

$$V = F = 2^{n-1}, \qquad E = n2^{n-2}, \qquad |\operatorname{Aut}| = n2^n, \qquad g = 1 + 2^{n-3}(n-4). \tag{2}$$

Its group is the semidirect product $E_n \rtimes D_{2n}$, where E_n is the even-coordinate subspace of $\mathbb{F}_2^n$. The proof in [C] uses the halving of the generalized cube, the two vertex colors, and the span of the edge differences. We use that established construction here.

The exact later source is [LZ25], arXiv:2505.09505v1, 14 May 2025, twelve PDF pages. Its Theorem 1.3 asserts the existence, type, self-duality and number of flags in (2). Its Theorem 3.4 supplies an explicit semidirect model and the presentation below. Corollaries 3.5 and 3.6 specialize the parameters. We recover these statements for their full parameter ranges. We do not assert a new existence theorem, a replacement for every auxiliary calculation in that paper, or a new solution of its higher-rank Conjecture 3.7. The latter is a separate question with separate counterexample evidence in this project.

Theorem 1 (polygon construction; exact presentation of Li–Zhang). For every integer $n \geq 3$, the group

$$\begin{aligned} P_n = \langle r_0, r_1, r_2 \mid\ & r_0^2 = r_1^2 = r_2^2 = (r_0 r_2)^2 = 1, \\ & (r_0 r_1)^n = (r_1 r_2)^n = 1, \\ & \big((r_1 r_2)^k (r_0 r_1)^k\big)^2 = 1 \quad (1 \leq k \leq n-1) \rangle \end{aligned} \tag{3}$$

is isomorphic, with the marking described below, to $E_n \rtimes D_{2n}$. This marking is a string C-group of type $\{n, n\}$, of order $n2^n$, and admits reversal by an involutory automorphism. Consequently it is the automorphism group of a self-dual regular polyhedron with $n2^n$ flags.

The existence, size and duality parts are immediate from the cited polygon construction. Sections 2 and 3 supply the marked identification and the universal presentation, so that the conclusion is stronger than an order calculation for a quotient of (3).

2. Coordinates and the distinguished marking

Let $\Omega = \mathbb{Z}/n\mathbb{Z}$, let e_i be the coordinate vectors in $M = \mathbb{F}_2^{\Omega}$, and put

$$E_n = \{v \in M : \textstyle\sum_{i\in\Omega} v_i = 0\}. \tag{4}$$

The dihedral group $H = \langle a, b \mid a^n = b^2 = 1,\ bab = a^{-1}\rangle$ acts on Ω by $a(i) = i+1$ and $b(i) = -i$. It acts on M by $ge_i = e_{g(i)}$, preserving E_n. Throughout, permutation products act on the rightmost factor first. We use the left semidirect multiplication

$$(v, g)(w, h) = (v + gw, gh). \tag{5}$$

Put $s = ab$, so that $s(i) = 1 - i$, and $q = e_0 + e_1$. In $G_n = E_n \rtimes H$, define

$$r_0 = (0, s), \qquad r_1 = (0, b), \qquad r_2 = (q, s). \tag{6}$$

These are nonidentity involutions, since $sq = q$. We have $r_0 r_2 = (q, 1)$ and $r_0 r_1 = (0, a)$.

The comparison with the source's matrices is also explicit. In the ordered basis

$$f_j = e_0 + e_{n-j} \qquad (1 \le j \le n-1) \tag{6a}$$

of E_n, we have $bf_j = f_{n-j}$, $af_1 = f_{n-1}$, and $af_j = f_{j-1} + f_{n-1}$ for $2 \le j \le n-1$. Thus the matrices of b, a are exactly the matrices V, A of [LZ25, Section 3], and the matrix of $s = ab$ is $U = AV$. The source's final coordinate vector is $f_{n-1} = q$. This identifies both its representation and its three marked generators, with the consistent multiplication (5).

To compare this marking with the earlier construction, take the standard distinguished generators of the generalized cube to be $(e_0, 1), (0, s), (0, b)$ in $M \rtimes H$. Halving replaces the first two by $(e_0, 1)(0, s)(e_0, 1) = (q, s)$ and $(0, s)$, retaining $(0, b)$. Ordered along the resulting three-node string and then reversed, these are exactly (6). Thus the even-subset construction is the polygon instance of [C], with an explicit identification of all distinguished generators.

The reversal is particularly simple in these coordinates. For $g \in H$, set

$$c(g) = e_0 + ge_0. \tag{7}$$

This belongs to E_n and satisfies $c(gh) = c(g) + gc(h)$. Consequently

$$\delta(v, g) = (v + c(g), g) \tag{8}$$

is an automorphism of G_n. It is conjugation by $(e_0, 1)$ in $M \rtimes H$, and $\delta^2 = 1$. Since $c(b) = 0$ and $c(s) = q$, it fixes r_1 and interchanges r_0, r_2. In particular both adjacent product orders are exactly n.

For completeness, the coordinate model also checks the exact intersection and generation statements needed by the presentation. The groups

$$A = \langle r_0, r_1\rangle = \{(0, g) : g \in H\}, \qquad B = \langle r_1, r_2\rangle = \delta(A) \tag{9}$$

meet in the stabilizer of 0 in H, namely $\langle r_1\rangle$. The third maximal marked subgroup is $C = \langle r_0, r_2\rangle = \{1, r_0, r_2, (q, 1)\}$. It has $A \cap C = \langle r_0\rangle$, since its other two elements have nonzero translation coordinate. Applying δ gives $B \cap C = \langle r_2\rangle$. Each of A, B, C is a rank-two C-group. Intersecting any two proper marked subgroups first inside these three maximal subgroups therefore proves (1); intersections involving the full group are tautological.

The generated group contains H and translation by q. Its conjugates include all $e_i + e_{i+1}$. The edges of the cycle span E_n: summing edges along a path from 0 to i gives $e_0 + e_i$, and these $n - 1$ vectors form a basis. Hence the three markers generate the whole G_n, whose order is $2^{n-1} \cdot 2n = n2^n$.

3. The universal presentation

Here we prove that (3) has no larger group above the displayed model. The commuting-involution argument is the one underlying [LZ25, proof of Theorem 3.4]; the normality and order steps are written out to give a closed proof.

In the abstract group (3), put

$$x = r_0 r_1, \qquad y = r_1 r_2, \qquad b = r_1, \qquad t_k = y^k x^k \quad (k \in \mathbb{Z}). \tag{10}$$

The relations $x^n = y^n = 1$ imply $t_{k+n} = t_k$, and the last line of (3) implies $t_k^2 = 1$ for every integer k, including $t_0 = 1$. Inverting t_k gives

$$y^k x^k = x^{-k} y^{-k}. \tag{11}$$

Conjugating this equality by x^k also gives $x^k y^k = y^{-k} x^{-k}$. For arbitrary $i, j \in \mathbb{Z}$, we obtain

$$\begin{aligned} t_i t_j &= x^{-i} y^{j-i} x^j \\ &= x^{-j} (x^{j-i} y^{j-i}) x^j \\ &= x^{-j} y^{i-j} x^i = t_j t_i. \end{aligned} \tag{12}$$

It follows that

$$N = \langle t_k : k \in \mathbb{Z} \rangle = \langle t_1, \dots, t_{n-1} \rangle \tag{13}$$

is an elementary abelian group of order at most 2^{n-1}.

We now check normality, before using any quotient notation. Direct multiplication gives

$$x^{-1} t_k x = t_1^{-1} t_{k+1}. \tag{14}$$

Thus $x^{-1} N x \subseteq N$; both subgroups have the same finite order, so equality holds. Since $bxb = x^{-1}$ and $byb = y^{-1}$, we also have $b t_k b = t_{-k}$. Hence b normalizes N. Finally $y = t_1 x^{-1}$, and $r_0 = xb, r_1 = b, r_2 = by$, so x, b, N generate the entire presented group. This proves $N \lhd P_n$.

Modulo N, we have $y = x^{-1}$. The quotient is generated by the images of x, b, with $x^n = b^2 = 1$ and $bxb = x^{-1}$; its order is at most $2n$. The normal subgroup and quotient are both finite, and therefore

$$|P_n| = |N| \, |P_n / N| \le 2^{n-1} \cdot 2n = n 2^n. \tag{15}$$

In the coordinate model, $x = (0, a)$, and $y = \delta(0, a^{-1})$. Thus

$$y^k x^k = (e_0 + e_{-k}, 1). \tag{16}$$

All of (3) hold, and (6) generates G_n, so there is a marked epimorphism $P_n \twoheadrightarrow G_n$. Section 2 gives $|G_n| = n 2^n$, which is the upper bound (15). The epimorphism is consequently an isomorphism. Moreover (16) identifies $t_1, \dots, t_{n-1}$ with a basis of E_n; no additional relation among these displacements is hidden in the upper bound. This proves Theorem 1. □

4. Parameter specializations and conventions

For every odd $n \ge 3$, Theorem 1 has exactly the type, number of flags and self-duality asserted in [LZ25, Corollary 3.5]. For integers $p \ge 2$, $s \ge 0$, putting $n = 2^s p \ge 3$ yields

$$|P_n| = 2^{2^s p + s} p, \tag{17}$$

which is the flag count in its Corollary 3.6. The latter's literal parameter range also includes $(p, s) = (2, 0)$. This is outside Theorem 1 but is supplied directly by C_2^3 on its three coordinate involutions: it is

a string C-group of type $\{2, 2\}$, has eight flags and admits reversal of the coordinates. We do not extend the faithful permutation action on an n-cycle to this exceptional value.

There is a convention issue in the source's displayed multiplication on physical page 7. It specifies a homomorphism $\varphi : H \to \mathrm{Aut}(E_n)$ and column vectors, but writes $(v, g)(w, h) = (v + \varphi(g)^{-1}w, gh)$. With ordinary left composition, inversion is an antihomomorphism and that formula is not generally associative. For example in the equivalent even-subset action with $n = 3$, put $u = (0, a)$, $v = (0, b)$, $w = (e_0 + e_1, 1)$. Under the inverse-action formula the translation coordinate of $(uv)w$ is $e_0 + e_1$, whereas that of $u(vw)$ is $e_1 + e_2$. Equation (5) fixes the convention. One may instead use a right action, provided the corresponding order of composition is changed consistently. Our recovery concerns the main group and polytope statements and the exact abstract presentation (3); it does not endorse the inconsistent displayed product as written.

5. Attribution, reuse and verification

The generalized-cube and halving construction, its polygon outputs and its duality belong to [MW20] and the antecedents attributed in [C]. The main later existence statement and presentation belong to [LZ25]. The displacement commutativity argument (12) is an expanded source argument. The regular-polytope correspondence is retained from [MS02]. The explicit common marking (6), the resulting source-to-source identification, the convention check and the formal verification are the work of this application. We assert no priority for an underlying new group-theoretic idea.

The mathematical dependency path is

$$\begin{aligned} &\text{generalized cubes and halving} \longrightarrow \text{[C, C.Polygon]} \longrightarrow \\ &\text{marked identification (6)} \longrightarrow \text{presentation (3) and its specializations.} \end{aligned} \tag{18}$$

The full relevant topic umbrella is charged once in a topic comparison. The later application pays for the identification, the complete presentation proof and its parameter adapters. Another topic pays its own umbrella acquisition. Formula (18) records actual reuse; it is not a compression ratio. No matching typeset cost or additional admitted source credit is assigned in this author packet before independent review.

The accompanying Lean file proves (12), (14), commutativity of the subgroup generated by all displacements, and the passage from the finite relations $1 \leq k < n$, $x^n = y^n = 1$, to all integer-indexed involutions and their commutative closure. It uses only standard kernel axioms. This is a proof of the algebraic core for arbitrary groups and all positive periods, not a finite sample. The full dimension calculation, normality, order bound, semidirect marking, geometric correspondence and source identification are written proofs here; this packet does not claim a complete Lean formalization of Theorem 1. A separate executable diagnostic checks the explicit model and all marked intersections for several small n, with an explicit failure of the inverse-action product. Those checks provide implementation checks, not a proof for unbounded n.

References

[C] *Halving and Petrial operations on regular geometries*, project common manuscript, frozen phase150, especially Section 5, Corollary C.Polygon and its preceding group calculation. The retained edition and its provenance are bound in the accompanying contract.

[LZ25] M. Li and W.-J. Zhang, *Regular 3-polytopes of type* $\{n, n\}$, arXiv:2505.09505v1, 14 May 2025. Theorem 1.3, Theorem 3.4 and Corollaries 3.5–3.6. https://arxiv.org/abs/2505.09505v1

[MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Section 2E.

[MW20] A. Montero and A. Ivić Weiss, *Locally spherical hypertopes from generalised cubes*, arXiv:2001.11082v1, 29 January 2020. https://arxiv.org/abs/2001.11082v1